Eco-Driving Control Architecture for Platoons of Uncertain Heterogeneous Nonlinear Connected Autonomous Electric Vehicles

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***Abstract*— The improvement of energy performance for platoons of autonomous connected vehicles is one of the major challenges the road transport sector is facing with. To this aim, this work addresses and solves the energy-consumption problem for uncertain heterogeneous electric nonlinear autonomous vehicles platoon via a novel Eco-Driving Control Architecture able to optimize its energy consumption performance while ensuring the fulfillment of the optimal leader tracking trajectory. Specifically, it consists of a Nonlinear Model Predictive Control (NMPC) strategy, driving the leader motion and computing the optimal ecological trajectory to be imposed on the whole platoon, and a novel distributed exponentially-stable robust PID-like protocol, driving the follower vehicles motion for achieving a precise leader-tracking with a desired transient behavior as required for the accurate implementation of the energy-saving control. The exponential stability of the overall vehicular network is analytically proven with the Lyapunov theory and the derived robust stability conditions allow the proper tuning of the control gains on the basis of the desired decay rate. The efficiency of the proposed approach is corroborated via the high-fidelity Mixed Traffic Simulator (MiTraS) co-simulation platform under different operative scenarios and a wide uncertainty range for the vehicles parameters. Simulation results confirm how the proposed architecture ensures the eco-driving behaviour for the whole vehicles platoon.**

***Index Terms*— Eco-drive, cooperative driving of heterogeneous autonomous connected vehicles, nonlinear uncertain vehicle platoon, electric vehicle, distributed robust exponential-stable PID, nonlinear model predictive control.**

I. Introduction

THE transportation sector is a major user of energy worldwide since the number of vehicle, as well as their usage, is increasing [1]. To face this problem, the main lines of activity are shifting transport towards the least polluting and most efficient modes, while employing more sustainable technologies and applications. Among them, Electric Vehicle (EV) is thought to be the most promising, efficient and viable solution to reduce both energy consumption and pollutant emissions produced by conventional vehicles [2]. However, despite the growth of EVs market deployment, sustainability and environmental benefits are usually behind consumption performance in the customers ranking [3]. Thus, the improvement of energy consumption performance is a crucial aspect to foster the transition to e-mobility.

In addition, Connected and Autonomous Vehicles (CAVs) have received a lot of attention due to their ability to optimally plan and execute driving operations through the usage of the surrounding road traffic environment information as obtained from communication and detection technologies [4], [5]. One crucial aspect to further enhance road efficiency, safety and energy-saving, is to cooperatively drive along the road by operating platoons of CAVs capable to reach and maintain a safe inter-vehicle distance while tracking, at the same time, desired speed and acceleration profiles [6], [7]. Indeed, experimental results have proved a $6\% - 10\%$ fuel economy improvement in platooning application due to the reduction of aerodynamic resistance forces and the minimization of unnecessary acceleration and deceleration manoeuvres [8]. Therefore, in recent years, research interests in energy performance have enormously increased and several different eco-driving motion control strategies have been proposed in technical literature. Global optimization control techniques, such as Dynamic Programming (DP) [9], minimum residual method [10] and Pontryagin's Principle [11], have been proposed for finding the optimal eco-driving speed profile to be imposed on the whole vehicles platoon. However, these kinds of centralized strategies are not practical in real traffic situations due to the heavy computational effort required [12]. Thereby, by leveraging the distributed control architecture concept, different control approaches have been suggested, such as the H_∞ energy-economy oriented controller in [13] that guarantees the asymptotic stability for homogeneous electric vehicles platoons, but not considering the leader-tracking performance issues. By exploiting, instead, dynamic optimization techniques in a distributed context, [14] proposes a hierarchical fuel-saving control architecture for homogeneous nonlinear Internal Combustion Engine (ICE) vehicles composed of a higher layer for route speed profile calculation and a lower layer for speed tracking. More specifically, the receding DP method is used for the optimization of the leading vehicle profile, which is tracked by the followers whose controller is designed according to the backstepping approach.

Conversely, to deal with multiple platooning control objectives (e.g. tracking, safety, energy economy and ride comfort) in a

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unified framework, while handling constrained on system state variable, the Model Predictive Control (MPC) approach has been proposed by the technical literature [15]. By leveraging this control framework, [16] suggests an eco-driving control scheme for homogeneous nonlinear platoons of plug-in hybrid electric vehicles, where the leading vehicle motion is driven by a Nonlinear MPC (NMPC) while the followers are equipped with a classical string-stable Cooperative Adaptive Cruise Controller (CACC) enforced with a predictive nonlinear reference governor whose aim is to keep the vehicle outputs within some specified constraints (such as maximum/minimum speed and acceleration). The more challenging problem of finding eco-driving control solution for heterogeneous nonlinear electric vehicles platoons is, instead, addressed in [17]. Herein the leading vehicle, equipped with a NMPC, is responsible for the speed profile optimization to be imposed to followers, while these latter are equipped with feed-forward control action compensating nonlinearities plus a diffusive state-feedback action for safely tracking the desired motion. However, degradation of energy-recovery performance occurs when the computation of optimal control input is demanded to static feedback control law.

Despite the increase in the computational effort required by the control architecture, the technical literature has also proposed Distributed MPC (DMPC) strategies which are proven to be more effective in ensuring the optimization of the leader speed profile to be tracked, of the energy consumption and leader tracking performance [12]. Along this line, [18] design, for homogeneous platoons of electric vehicles, an ecological CACC based on DNMPC, whose cost function includes, for each vehicle, the speed of the follower vehicle, the range deviation and the energy consumption. Moreover, authors in [19] develop a two-layer look-ahead control architecture for heavy-duty nonlinear homogeneous/heterogeneous ICE vehicles platoon where DP has been used as centralized coordinator to compute an off-line fuel-saving speed profile (based on the look-ahead information) for the whole platoon, while a DMPC has been used for the real-time motion control of each vehicle. The proposed motion control approach only ensures bounded stability with collision avoidance requirement but does not guarantee satisfactory tracking accuracy. Again, the eco-cooperative look-ahead control problem for a platoon of gasoline heterogeneous nonlinear vehicles traveling on an extra-urban road with varying slopes is addressed in [20]. The proposed DMPC has been designed taking into account rotational inertia coefficient related to a reduction ratio of gearbox, aerodynamic drag and model constraints, while a particle swarm optimization algorithm has been embedded so to quickly solve the online optimization problem. However, it is worthy highlighting that the multi-layer look-ahead control architecture is not a suitable solution for practical real-time heterogeneous vehicle, especially when look-ahead information are not available, or vehicle dynamics are uncertain. Indeed, the robustness concerning uncertain nonlinear dynamics is crucial when deploying cooperative driving applications since many mismatches between the actual plant and its control-oriented model inevitably arise due to different environmental conditions, parameter variations, and unmodeled dynamics [21]. Moreover, these uncertainties strongly affect the energy consumption evaluation [22], hence they need to be considered for the designing of robust eco-driving control strategies. A first attempt along this direction is given by the very recent work [23] that solves this problem for ICE vehicles platoon via a double-layer control architecture. The centralized upper-layer aims at computing the fuel-optimal feasible speed profile based on the preview information on road condition and all vehicles- belonging to the platoon- characteristics through the dynamic programming approach. The distributed lower-layer, leveraging a potential function-based method, guarantees the achievement of speed tracking and collision avoidance among vehicles, ensuring the convergence of the speed tracking error and uniform boundedness of the spacing errors. However, the selection of the optimized feasible speed profile in a centralized fashion, as well as the exploitation of dynamic programming techniques, are not a suitable solution for real driving applications due to its heavy computational burden [12], [24].

Motivated by the above discussion, in this work we propose a novel distributed eco-driving control architecture for platoons of uncertain heterogeneous nonlinear electric CAVs able to optimize the energy consumption performance while ensuring the fulfillment of the optimal leader trajectory tracking purpose in a safe manner. The design is made, differently from the technical literature in the field, by leveraging a more realistic representation of heterogeneous nonlinear vehicle dynamics which are affected by time-varying nonlinear uncertainties and also account for the air-drag variation due to platooning effect (usually neglected in the different control-oriented models proposed in the technical literature on heterogeneous and nonlinear platoons). The designed control architecture consists of: i) a NMPC strategy driving the leader motion which, given a desired reference behaviour provided by the RSU via V2I paradigm, is responsible for computing an optimal ecological trajectory to be imposed on the whole vehicles within the platoon; ii) a novel robust exponential-stable PID-like strategy for driving the follower vehicles motion in order to achieve a precise leader-tracking with a desired transient behaviour and prescribed decay rate, despite the presence of unknown time-varying uncertainties affecting their dynamics. Notably, differently from the technical literature in the field (e.g. [16], [17], [23], [25]), the proposed eco-driving approach explicitly takes into account the air-drag reduction effects and the regenerative braking ability during acceleration and deceleration phases. Regarding the second strategy, it plays a key role for our energy saving control objective since a precise and exponential convergence of the speed error with desired transient behaviour is required to ensure the accurate implementation of the ecological control logic; differently from the above mentioned technical literature which only ensures the asymptotic stability. By combining the advantages of the aforementioned control methods, the proposed eco-driving architecture brings a significant energy consumption saving for the overall autonomous vehicles platoon. The exponential stability of the novel distributed PID-like controller is proven via the Lyapunov theory and the derived stability conditions are expressed as a set of feasible LMIs. These latter, given a

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desired decay rate for tracking, allow the proper tuning of the control gains so to ensure the desired transient performances and the convergence time for the longitudinal closed-loop trajectories. Moreover an explicit expression of the bound state trajectories as function of the decay rate is provided. Theoretical results are then validated via the virtual testing simulation platform MiTRaS [26] that allows an extensive evaluation of the proposed eco-driving control architecture due to: the possibility to control all the aspects of the test; the possibility to assess of the performance of autonomous vehicles in a wide-range of realistic traffic scenarios; the repeatability of test by varying the unknown vehicles parameters over a wide range of values; the replicability of each test without deviations due to stochastic phenomena and faults in the functioning of a system; the deeply evaluation and validation of the safety principles. Virtual simulations, carried in different operative conditions, confirm the effectiveness of the proposed control eco-driving control architecture in achieving both energy-saving and leader-tracking control objectives.

Finally, the rest of this paper is organized as follows. Notation and math preliminaries are both introduced in Section II. In section III the problem statement is presented, while the explanation of the proposed eco-driving control architecture, the derivation of the closed-loop vehicular network dynamics and the proof of asymptotic stability of the control strategy are presented in Section VI. The performance analysis is introduced in Section V, and related results are disclosed in Section VI. Conclusion are finally drawn in Section VII.

II. Math Preliminaries

The Vehicle-to-Vehicle (V2V) wireless communication network, allowing the information sharing among N vehicles, can be modeled as a directed graph (or digraph) $\mathscr{G}_N = (\mathscr{V}, \mathscr{E}, \mathscr{A})$ of order N where $\mathscr{V} = \{1, \ldots, N\}$ and $\mathscr{E} \subseteq \mathscr{V} \times \mathscr{V}$ indicate the set of vehicles and the set of communication links, respectively. The graph topology is described by the adjacency matrix $\mathscr{A} = [a_{ij}] \in \mathbb{R}^{N\times N}$, whose elements are such that $a_{ij} = 1$ if the vehicle j communicates with vehicle i (but not necessarily viceversa), $a_{ij} = 0$ otherwise. Note that, we assume that the self-loops are not present, i.e. $a_{ii} = 0, \ \forall i \in \mathcal{G}_N$. Thus, it is possible to introduce the in-degree matrix $\mathscr{D} = diag(d_1, d_2, \cdots, d_N) \in \mathbb{R}^{N\times N}$, whose elements $d_i = \sum_{j=1}^{N} a_{ij}$. Accordingly, the Laplacian matrix can be defined as $\mathscr{L} = \mathscr{D} - \mathscr{A}$. In order to model the platoon of N autonomous vehicles plus a leader, here we exploit an augmented directed graph $\mathscr{G}_{N+1}$, where the additional agent, defined as leader and labeled with 0, imposes the reference behaviour to the whole platoon. To model the resulting graph $\mathscr{G}_{N+1}$, we also leverage the pinning matrix [27], [28] $\mathscr{P} = \text{diag}\{p_1, p_2, \ldots, p_N\}$, such that $p_i = 1$ when the leader information is directly available for the $i-th$ agent/vehicle, 0 otherwise. In so doing, the communication graph is described via the matrix $\mathscr{H} = \mathscr{L} + \mathscr{P} \in \mathbb{R}^{N\times N}$. Moreover, according to technical literature [21] we assume the following assumption to be held.

Assumption 1 ([28]): The communication topology among the vehicles contains a directed spanning tree with the leader as root.

Indeed, the typical network topologies for vehicle platoons, including Predecessor-Follower (P-F), Leader-Predecessor-Follower (L-P-F) and Broadcast (BR) satisfy Assumption 1. Finally, the following lemmas will be useful in the stability analysis.

Lemma 1 ([27]): If $\mathscr{G}_{N+1}$ satisfies Assumption 1, then the matrix $\mathscr{H} = \mathscr{L} + \mathscr{P} \in \mathbb{R}^{N\times N}$ is positive definite.

Lemma 2 ([29]): Given a generic matrix $F \in \mathbb{R}^{n\times n}$ and a vector $x \in \mathbb{R}^{n\times 1}$, F and $\frac{1}{2}(F + F^\top)$ both generate the same quadratic form, i.e. the following relation holds: $x^\top F x = \frac{1}{2}x^\top(F + F^\top)x$, where $\frac{1}{2}(F + F^\top)$ is a symmetric matrix.

Lemma 3 ([30]): Suppose that Assumption 1 holds. Then, there exist a positive vector $\theta = [\theta_1, \ \theta_2, \ \cdots, \ \theta_N]^\top$ such that

$$\mathscr{H}\theta = 1_N \tag{1a}$$

$$\Theta\mathscr{H} + \mathscr{H}^\top\Theta > 0 \tag{1b}$$

where $\Theta = diag(1/\theta_1, 1/\theta_2, \cdots, 1/\theta_N)$ and $\mathcal{H}$ is an M-matrix according to Lemma 1.

III. Problem Statement

Consider a platoon composed of $N + 1$ Connected Autonomous Electric Vehicles (CAEVs) with a regenerative braking system where the leader of the fleet is indexed with 0. Each vehicle is assumed to be equipped with on-board inertial sensors and GPS receiver to measure its state information (e.g. its absolute position, speed and acceleration) that are shared with all the other vehicles within the platoon via a V2V communication network. Moreover, CAEVs are connected with a Road Side Unit (RSU) via Vehicle-to-Infrastructure (V2I) paradigm which suggests to the whole platoon the reference speed profile (computed by the RSU itself with the aim at minimizing the total freeway travel time of the appraised road section (see e.g. [31])) to be tracked along the road section in which CAEVs travel.

Based on the reference behaviour suggested by RSU, our aim is to design a novel eco-driving control architecture able to guarantee an ecological and safe platooning movement for a group of CAEVs. More specifically, the eco-driving control architecture is designed to fulfill the following objectives: i) to compute an optimal speed profile able to guarantee energy saving; ii) to guarantee that all the CAEVs members drive with the computed ecological speed profile while maintaining a desired inter-vehicle distance; iii) to ensure robustness w.r.t. uncertainties acting on the vehicle dynamics. In so doing, the proposed eco-driving control architecture not only allows achieving platooning formation, but also ensures energy saving for the whole vehicular platoon. Note that, without loss of generality, our eco-driving architecture can be exploited even if the reference behaviour is not provided by the RSU, but, instead, by another vehicle, non member of the platoon, travelling ahead of the first one (see e.g. [17]).

A. CAEVs Longitudinal Dynamics

The behaviour of CAEVs is described by their longitudinal dynamics which are inherently nonlinear, heterogeneous, uncertain and mainly governed by the drive-train

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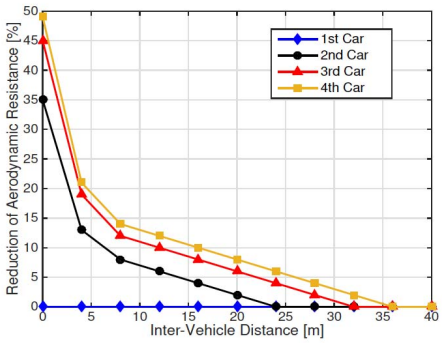

Fig. 1. Air-drag reduction of passenger cars [36].

dynamics, including aerodynamics drag, rolling resistance and gravitational force [32]. The motion of the i-th CAEV ($\forall i = 1, \cdots, N$) is described by the following nonlinear system [33], [34]:

$$\begin{aligned}\dot{p}_i(t) &= v_i(t)\\ \dot{v}_i(t) &= \frac{\eta_i(t)}{R_i m_i(t)}u_i(t) - g sin(\theta_i(t)) - g\mu_i(t)cos(\theta_i(t))\\ &\quad - \frac{\rho_{air}}{2m_i(t)}C_{D_i}(t)(1-\phi_i(p_i(t), p_{i-1}(t)))A_{f_i}(t)v_i^2(t),\end{aligned} \tag{2}$$

where $p_i(t)$ $[m]$ and $v_i(t)$ $[m/s]$ are the position and the speed of the $i-$th CAEV, respectively; $u_i(t)$ $[Nm]$ is the control input representing the driving/braking torque; $m_i(t)$ $[kg]$ is the vehicle mass; $\eta_i(t)$ is the drive-train mechanical efficiency; R_i $[m]$ is the wheel radius; $\mu_i(t)$ is the rolling resistance coefficient, computed as $\frac{C_r(t)}{1000}(c_1(t)v_i(t)+c_2(t))$ [35], being $C_r(t)$, $c_1(t)$ and $c_2(t)$ the rolling resistance parameters; ρ_{air} $[kg/m^3]$ is the air density; $C_{D_i}(t)$ is the air-drag coefficient; $A_{f_i}(t)$ $[m^2]$ is the frontal area of the i-th CAEV; g $[m/s^2]$ is the gravity acceleration while $\theta_i(t)$ $[rad]$ is the road-track slope. The air-drag reduction factor $\phi_i(p_i(t), p_{i-1}(t))$, computed according to [36], relies on both inter-vehicle distance between adjacent vehicles and on the specific position of the vehicle i within the platoon. More specifically, as depicted in Fig. 1, the first vehicle (the leader in this case) has zero reduction of air-drag resistance coefficient, while it increases as the positioning index of a vehicle within the platoon increases and, in the end, for all positions $n \geq 4$ it assumes the same value as for the 4th vehicle. The value of the vehicle parameters strongly depends on the different driving conditions [21] and the change of the operating speed [37]. Here, in order to take into account this variability, we assume the parameters to be uncertain and time-varying, i.e. we model it as follows:

$$\begin{aligned}m_i(t) &= \bar{m}_i + \delta m_i(t), \quad \eta_i(t) = \bar{\eta}_i + \delta\eta_i(t),\\ C_{Di}(t) &= \bar{C}_{Di} + \delta C_{Di}(t), \quad \mu_i(t) = \bar{\mu}_i + \delta\mu_i(t)\\ A_{fi}(t) &= \bar{A}_{fi} + \delta A_{fi}(t),\end{aligned} \tag{3}$$

where the notation $\bar{(\cdot)}_i$ indicates the nominal value of the vehicle parameters, assumed to be known, while $\delta(\cdot)_i(t)$ represent their corresponding time-varying uncertainty. The uncertainties are such that the following assumption holds.

Assumption 2 ([21], [37]): For each vehicle i within the fleet, the time-varying parameter uncertainties are bounded with a known upper-bound value, i.e. $\|\delta_i(\cdot)\| \leq \bar{\delta}_i(\cdot) < +\infty, \quad \forall i$.

Now, by naming with $x_i(t) = [p_i(t), v_i(t)]^\top \in \mathbb{R}^{2\times 1}$ the i-th vehicle state vector, according to the uncertainty decomposition in (3), the nonlinear dynamics in (2) can be rewritten in a more compact form as

$$\begin{aligned}\dot{x}_i(t) &= Ax_i(t) + B(\varphi_i(v_i(t)) + \Delta\varphi_i(v_i(t)))\\ &\quad + B(\bar{b}_i + \Delta b_i(t))u_i(t),\end{aligned} \tag{4}$$

where

$$A \in \mathbb{R}^{2\times 2} = \begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}, \quad B \in \mathbb{R}^2 = \begin{bmatrix}0\\ 1\end{bmatrix},$$

$$\begin{aligned}\varphi_i(v_i(t)) &= -\frac{\rho_{air}\bar{C}_{Di}\bar{A}_{fi}}{2\bar{m}_i}(1-\phi_i(p_i(t), p_{i-1}(t)))v_i^2(t)\\ &\quad - g sin(\theta_i(t)) - g\bar{\mu}_i cos(\theta_i(t)),\end{aligned}$$

$$\begin{aligned}\Delta\varphi_i(v_i(t)) &= -\frac{\rho_{air}(1-\phi_i(p_i(t), p_{i-1}(t)))}{2}\Big(-\frac{\bar{C}_{Di}\bar{A}_{fi}\delta m_i(t)}{\bar{m}_i(\bar{m}_i+\delta m_i(t))}\\ &\quad + \frac{\bar{C}_{Di}\delta A_{fi}(t)+\bar{A}_{fi}\delta C_{Di}(t)+\delta C_{Di}(t)\delta A_{fi}(t)}{\bar{m}_i+\delta m_i(t)}\Big)v_i^2(t)\\ &\quad - g\delta\mu_i(t)cos(\theta_i(t))\end{aligned}$$

$$\bar{b}_i = \frac{\bar{\eta}_i}{R_i\bar{m}_i},$$

$$\Delta b_i(t) = \frac{R_i\bar{m}_i(\bar{\eta}_i + \delta\eta_i(t)) - R_i\bar{\eta}_i(\bar{m}_i + \delta m_i(t))}{R_i^2\bar{m}_i(\bar{m}_i + \delta m_i(t))}. \tag{5}$$

Note that, according Assumption 2, it also follows that $\|\Delta\varphi_i(v_i(t))\| \leq \bar{\Delta}\varphi_i < +\infty$ and $\|\Delta b_i(t)\| \leq \bar{\Delta}b_i < +\infty$, $(i = 1, 2, \cdots, N)$.
Similarly, the leader dynamics are described by the following nonlinear system:

$$\begin{aligned}\dot{x}_0(t) &= Ax_0(t) + B(\varphi_0(v_0(t)) + \Delta\varphi_0(v_0(t)))\\ &\quad + B(\bar{b}_0 + \Delta b_0(t))u_0(t),\end{aligned} \tag{6}$$

with $x_0(t) = [p_0(t) \;\; v_0(t)]^\top \in \mathbb{R}^2$, being $p_0(t)$ $[m]$ and $v_0(t)[m/s]$ the position and the velocity of the leading vehicle, respectively; $u_0(t)$ $[Nm]$ is the leader control input that represents the vehicle propulsion torque.

B. CAEVs Battery Pack Model

The battery pack of each vehicle j ($j = 0, \cdots, N$) is modeled via an equivalent simplified electric circuit as in [38]. It is composed by an internal voltage source $E_{oc,j}$ $[V]$, two ideal diodes, and two inner resistances $R^+_{in,j}$ $[\Omega]$ and $R^-_{in,j}$ $[\Omega]$ representing the internal discharging and charging resistances of the battery. The value of these latter parameters depend on the current value of the battery State Of Charge (SOC) and are implemented as look-up tables [38]. The voltage at the terminal of the battery is computed as:

$$V_{t,j}(t) = \begin{cases}E_{oc,j} - R^+_{in,j}I_{bat,j}(t) & if\ discharging,\\ E_{oc,j} - R^-_{in,j}I_{bat,j}(t) & if\ charging.\end{cases} \tag{7}$$

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Indicating with $P_{bat,j}(t)$ $[W]$ the power required at the battery, the current $I_{bat,j}(t)$ $[A]$ can be derived as

$$I_{bat,j}(t) = \begin{cases} \dfrac{E_{oc,j} - \sqrt{E_{oc,j}^2 - 4\ R_{in,j}^{+} P_{bat,j}(t)}}{2R_{in,j}^{-}} & if\ discharging \\ \dfrac{E_{oc,j} - \sqrt{E_{oc,j}^2 - 4R_{in,j}^{-} P_{bat,j}(t)}}{2R_{in,j}^{+}} & if\ charging. \end{cases} \tag{8}$$

The battery State of Charge can be, hence, computed as

$$SOC_j(t) = \begin{cases} -\frac{1}{C_{bat,j}} \int_0^t I_{bat,j}(\tau) d\tau & if\ discharging \\ -\frac{\eta_{bat,j}}{C_{bat,j}} \int_0^t I_{bat,j}(\tau) d\tau & if\ charging. \end{cases} \tag{9}$$

where $C_{bat,j}$ $[Ah]$ is the battery capacity while $\eta_{bat,j}$ is the recharging efficiency.

C. Power-Based Energy Consumption Estimation Model

To estimate the energy consumption of each CAEV j ($j = 0, 1, 2, \cdots, N$), we exploit the Comprehensive Power-based EV Energy consumption Model (CPEM) [39]. It requires the CAEVs instantaneous speed, i.e. $\upsilon_j(t)$, and their characteristics as inputs, while the outputs are the instantaneous power consumed $P_{t,j}(t)$ $[W]$ and the total energy consumption (EC_j) $[kWh/km]$ for a specific driving cycle. To derive EC_j, we first compute the required electric power at wheels $P_{w,j}(t)$ $[W]$ for each vehicle j as:

$$P_{w,j}(t) = \Big(m_j a_j(t) + R_{g,j}(t) + R_{a,j}(t) + R_{r,j}(t)\Big)\upsilon_j(t), \tag{10}$$

where $a_j(t)$ $[m/s^2]$ is the acceleration; $R_{g,j}(t) = m_j g sin\theta_j(t)$ is the grade resistance; $R_{a,j}(t) = 0.5\rho_{air} A_{f_j} C_{D_j}(1 - \phi_j(p_j(t), p_{j-1}(t)))\upsilon_j^2(t)$ is the aerodynamic drag resistance; $R_{r,j}(t) = m_j g \cos\theta_j(t)\mu_i(t)$ is the rolling resistance. The electric motor power loss $P_{loss,j}$ $[W]$ referring to iron loss, copper loss and line loss inside the motor can be computed $\forall j$ as follows [40]:

$$P_{loss,j}(t) = \frac{r_j\ R_j^2}{K_j^2}\Big(m_j a_j(t) + R_{g,j}(t) + R_{a,j}(t) + R_{r,j}(t)\Big)^2, \tag{11}$$

where r_j $[\Omega]$ is the resistance of the motor armature; R_j $[m]$ is the radius of the wheel; $K_j = K_{a,j} \cdot \phi_{d,j} \cdot I_j$, being $K_{a,j}$ a constant of the armature, $\phi_{d,j}$ (in weber) the magnetic flux and I_j $[A]$ the current of the motor. Since vehicles are equipped with regenerative braking system, the power required to provide traction $P_{t,j}(t)$ $[W]$ can be expressed $\forall j$ as

$$P_{t,j}(t) = \begin{cases} \dfrac{\big(P_{w,j}(t) + P_{loss,j}(t)\big)}{\eta_i \cdot \eta_b}, & \text{if } P_{w,j}(t) \geq 0 \\ \big(P_{w,j}(t) + P_{loss,j}(t)\big) \cdot \eta_i \cdot \eta_b, & \text{if } P_{w,j}(t) < 0 \end{cases}$$

where η_b is the battery efficiency, while $\eta_{rb,j}(t)$ is the regenerative braking energy efficiency computed as:

$$\eta_{rb_j}(t) = \begin{cases} \left[e^{\frac{0.0411}{|a_j(t)|}}\right]^{-1} & if\ a_j(t) < 0, \\ 0 & if\ a_j(t) \geq 0. \end{cases} \tag{12}$$

Finally, considering also the electric power consumed by the auxiliary systems, i.e. $P_{aux,j} = 700$ $[W]$, the instantaneous total power required at battery $P_{bat,j}(t)$ $[W]$ for each CAEV j can be estimated as

$$P_{bat,j}(t) = P_{t,j}(t) + P_{aux,j}. \tag{13}$$

Finally, letting $d_{tot,j}$ $[km]$ the total travelled distance by the vehicle j during a driving cycle and given $P_{bat,j}(t)$, the average energy consumption EC_j $[kW/h\ km]$ can be computed as

$$EC_j = \frac{1}{3600000} \cdot \frac{1}{d_{tot,j}} \cdot \int_0^t P_{bat,j}(\tau)\, d\tau. \tag{14}$$

Now, the Eco-Driving Control Architecture design problem can be formulated as follows.

Problem 1: Consider a platoon of $N+1$ CAEVs sharing information with each other via V2V communication paradigm and with an RSU via V2I paradigm. Propose an Eco-Driving control architecture able to fulfill, despite the presence of uncertainties acting on the vehicle dynamics, the following objectives: i) computing an optimal speed profile to minimize the energy consumption of the leader vehicle by avoiding aggressive driving operations and achieving improved battery management, i.e. find $u_0(t)$ in (6) such that

$$\upsilon_0(t) = \underset{\upsilon_0}{\text{argmin}}\ P_{bat,0}(t); \tag{15}$$

ii) ensuring that all CAEVs within the platoon drive with the ecological speed profile $\upsilon_0(t)$ while moving in formation with a prescribed decay rate, i.e. find a distributed control protocol $u_i(t)$ in (4) such that $\forall i = 1, \cdots, N$

$$\lim_{t\to\infty} \|p_i(t) - p_j(t) - d_{ij}\| = 0, \quad \forall (i, j) \in \mathcal{E}_{N+1}, \\ \lim_{t\to\infty} \|\upsilon_i(t) - \upsilon_0(t)\| = 0, \tag{16}$$

being d_{ij} $[m]$ the desired inter-vehicle distance between the vehicle i and the vehicle j.

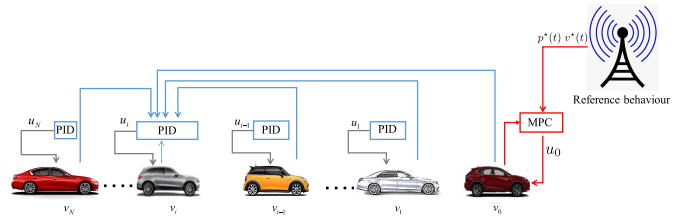

Fig. 2. Eco-driving control architecture.

IV. Eco-Driving Control Architecture

To solve Problem 1, we design the Eco-Driving control architecture schematized in Fig. 2. It is composed by: *a*) a NMPC strategy for driving the leader motion in order to achieve the control objective Problem 1-*i*; *b*) a robust exponential-stable PID-like strategy for driving the follower vehicles motion in order to reach the control objective Problem 1-*ii* with a prescribed decay rate. By combining the advantages of the aforementioned control methods, the proposed Eco-Driving architecture brings a significant energy consumption saving for the overall autonomous vehicles platoon.

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A. Design of the MPC Strategy for the Leading Vehicle

Given a desired trajectory imposed by RSU, i.e. $x_{ref}(t) = [p^\star(t),\ v^\star(t)]^\top$, the leading vehicle is responsible for generating and tracking an optimal energy-saving oriented reference behaviour to be also imposed to the whole platoon. To this end, we design the eco-driving control input $u_0(t)$ acting on the leader dynamics in (6) via a NMPC strategy. In this control framework, the motor/braking torque is computed as the solution of a constrained multi-objectives optimal control problem over a finite time horizon (named prediction horizon), and this procedure is repeated using updated state variables in a feedback way at every time step. Thereby, the leader control input design is formulated as the following constrained optimization problem:

$$\begin{aligned}
&\min_{u_0} \mathscr{J} = \int_t^{t+T} L(x_0(\tau,t), x_{ref}(\tau,t), u_0(\tau,t))d\tau \\
&\text{subject to} \\
&\qquad \dot{x}_0 = f_0(x_0, u_0) \\
&\qquad v_{0,\min} \le v_0(\tau,t) \le v_{0,\max} \\
&\qquad a_{0,\min} \le a_0(\tau,t) \le a_{0,\max} \\
&\qquad u_{0,\min} \le u_0(\tau,t) \le u_{0,\max}
\end{aligned} \tag{17}$$

where T $[s]$ is the prediction horizon duration; $x_0(\tau|t)$ and $u_0(\tau|t)$ are the prediction of the leader state and the control input trajectories, respectively, along the τ-axis starting from $x_0(t)$ as the initial state at $\tau = t$ over the prediction horizon T; the subscripts $_{\min}$ and $_{\max}$ denote the minimum and maximum bounds of the corresponding variable for the leading vehicle. The cost function L in (18) is designed as:

$$L = \omega_1 L_1 + \omega_2 L_2 + \omega_3 L_3 + \omega_4 L_4 + \omega_5 L_5 + \omega_6 L_6, \tag{18}$$

being $\omega_1, \cdots, \omega_6$ positive weights to be properly selected and

$$\begin{aligned}
&L_1 = \frac{1}{2}(v^\star(t) - v_0(t))^2, \quad L_2 = \frac{1}{2}P_{bat,0}^2(t) \\
&L_3 = \frac{1}{2}(p^\star(t) - p_0(t) - d_{ref,0})^2, \quad L_4 = \frac{1}{2}u_0^2(t), \\
&L_5 = e^{v_0(t) - v_{limit}(t)}, \\
&L_6 = \frac{1}{2}(a_0(t) + g\sin\theta(t))^2
\end{aligned} \tag{19}$$

where $d_{ref,0}$ $[m]$ is the desired distance w.r.t. to the desired reference position $p^\star$ $[m]$; $v_{limit}(t)$ $[m/s]$ represents the time-varying legal speed limit imposed on the road where the leader is moving along; $\theta(t)$ $[rad]$ is the slope of the road where vehicle is moving along.

Each term in (19) represents a specific control aim, namely: L_1 guarantees that the leading vehicle tracks the desired reference speed $v^\star(t)$; L_2 ensures the minimization of the instantaneous power consumption; L_3 ensures a pre-fixed spacing policy between the leader and the imposed reference position $p^\star(t)$; L_4 contains strong acceleration and deceleration manoeuvres, hence improving the driving comfort [25]; L_5 ensures that the leader speed does not exceed the legal speed $v_{limit}(t)$ imposed on the current road stretch; L_6 improves the driving experience by guaranteeing that vehicle increases its speed before entering an elevated slope section of the roads while controlling the deceleration levels when at downhill [41].

B. Design of Distributed Robust PID Control Strategy for Followers

To guarantee that each vehicle i ($\forall i = 1, \cdots, N$) tracks the optimized eco-driving reference behaviour computed by the leading vehicle, we design the control input $u_i(t)$ in (4) as the following robust exponentially stable Proportional-Integral-Derivative (PID) protocol:

$$\begin{aligned}
u_i(t) = &-\alpha\bar{b}_i^{-1}K_P\sum_{j=0}^{N}a_{ij}(p_i(t) - p_j(t) - d_{ij}) \\
&-\alpha\bar{b}_i^{-1}K_I\sum_{j=0}^{N}a_{ij}\int_0^t(p_i(s) - p_j(s) - d_{ij})ds \\
&-\alpha\bar{b}_i^{-1}K_D\sum_{j=0}^{N}a_{ij}(v_i(t) - v_j(t)),
\end{aligned} \tag{20}$$

where $\alpha \in \mathbb{R}_+$ is a tuning parameter; K_P and $K_D \in \mathbb{R}$ are the proportional gains, while K_I is the integral one; a_{ij} models the communication topology according to Section II. In what follows we derive exponential stability conditions for the control protocol (20) and the tuning of the control gains is provided.

1) Heterogeneous Uncertain Vehicular Platoon Closed-Loop Dynamics: To derive exponential stability conditions for the control protocol (20), we first derive the closed-loop dynamics of the vehicles platoon. To this aim, define the state error between the CAEV i and the leader 0 as

$$\tilde{x}_i(t) = x_i(t) - x_0(t) - d_i^\star \tag{21}$$

where $d_i^\star = [d_{i0}, 0]^\top \in \mathbb{R}^2$ being $d_{i0} \in \mathbb{R}_+$ the desired inter-vehicle distance between $i-$th vehicle itself and the leader. Given the dynamics of the follower and the leader as in (4)-(6) the error dynamics can be derived as

$$\dot{\tilde{x}}_i(t) = A\tilde{x}_i(t) + B\tilde{\Delta}f_i(v_i(t), v_0(t)) + B(\bar{b}_i + \Delta b_i(t))u_i(t), \tag{22}$$

being $\tilde{\Delta}f_i(v_i(t), v_0(t)) = (\varphi_i(v_i(t)) - \varphi_0(v_0(t))) + (\Delta\varphi_i(v_i(t)) - \Delta\varphi_0(v_0(t))) - Bb_0(t)u_0(t) \in \mathbb{R}$ such that, according Assumption 2, $\|\tilde{\Delta}f_i(v_i(t), x_0(t))\| \le \bar{\Delta}f_i < +\infty$, $(i = 1, 2, \cdots, N)$. Now, introducing the extended state vector $\xi_i(t) = [\tilde{x}_i(t), z_i(t)]^\top$, being $z_i(t) = \int_0^t(p_i(s) - p_j(s) - d_{i0})ds = l^\top\int_0^t\tilde{x}_i(s)ds$, with $l^\top = [1, 0] \in \mathbb{R}^{1\times 2}$, the closed-loop dynamics for the i-th CAEV in (22) can be recast as

$$\begin{aligned}
\dot{\xi}_i(t) = &\bar{A}\xi(t) + \bar{B}\tilde{\Delta}f_i(v_i(t), v_0(t)) \\
&- \bar{B}(\bar{b}_i + \Delta b_i(t))\big(\alpha\bar{b}_i^{-1}K^\top\sum_{j=0}^{N}a_{ij}(\xi_i(t) - \xi_j(t))\big),
\end{aligned} \tag{23}$$

being $\bar{A} = \begin{bmatrix} A & 0_{2\times 1} \\ l^\top & 0 \end{bmatrix}$, $\bar{B} = \begin{bmatrix} B \\ 0 \end{bmatrix}$, $K^\top = \begin{bmatrix} K_P & K_D & K_I \end{bmatrix}$. Finally, by introducing the global vectors $\xi(t) = [\xi_1^\top(t), \xi_2^\top(t), \cdots, \xi_N^\top(t)]^\top \in \mathbb{R}^{3N\times 1}$, $\tilde{\Delta}f(t) = [\tilde{\Delta}f_1(v_1(t), v_0(t)), \tilde{\Delta}f_2(v_2(t), v_0(t)), \cdots, \tilde{\Delta}f_N(v_N(t), v_0(t))]^\top \in \mathbb{R}^{N\times 1}$, $u(t) = [u_1(t), u_2(t), \cdots,$

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$u_N(t)]^\top \in \mathbb{R}^{N\times 1}$, and the matrices $\bar{b} = diag(\bar{b}_1, \bar{b}_2, \cdots, \bar{b}_N) \in \mathbb{R}^{N\times N}$, $\bar{\Delta} b(t) = diag(\Delta b_1(t), \Delta b_2(t), \cdots, \Delta b_N(t)) \in \mathbb{R}^{N\times N}$, after some mathematical steps, the closed-loop dynamics of the overall heterogeneous uncertain platoon can be written as

$$\dot{\xi}(t) = (I_N \otimes \bar{A})\xi(t) + (I_N \otimes \bar{B})\tilde{\Delta} f(t) - \alpha(\mathscr{H} \otimes \bar{B}K^\top)\xi(t) - \alpha(I_N \otimes \bar{B})\bar{\Delta} b(t)\bar{b}^{-1}(\mathscr{H} \otimes K^\top)\xi(t), \tag{24}$$

where the matrix $\mathcal{H} = \mathcal{L} + \mathcal{P} \in \mathbb{R}^{N\times N}$, being $\mathcal{L}$ and $\mathcal{P}$ the Laplacian and the Pinning matrix, respectively, defined according to Section II.

2) Stability Analysis and Control Gains Tuning: Now the the robust exponential stability conditions [42], as well as the control gains tuning, are provided by the following Theorem.

Theorem 1: Consider the uncertain nonlinear heterogeneous closed-loop platoon dynamics defined in (24) *such that Assumption 2 holds. Let the communication topology $\mathcal{G}_{N+1}$ be such that Assumption 1 holds. Given a decay rate β, if there exist scalars α, c, γ, a positive vector $\theta \in \mathbb{R}^{N\times 1}$ and a matrix Θ as in* (1a)-(1b)*, a symmetric positive definite matrices $P \in \mathbb{R}^{3\times 3}$, $S \in \mathbb{R}^{3N\times 3N}$ and some free matrices $M_k \in \mathbb{R}^{3N\times 3N}$, $k = 1, 2, \cdots, m$, with $m \in \mathbb{N}_{>0}$, such that*:

$$c < \underline{\lambda}_\Theta \theta_0, \tag{25}$$

$$\begin{bmatrix} \Psi & \Theta \otimes \bar{B} \\ \star & -\gamma^2 I_N \end{bmatrix} < 0, \tag{26}$$

where $\underline{\lambda}_\Theta$ is the minimum eigenvalue of the matrix $\Theta\mathscr{H} + \mathcal{H}^\top\Theta$, θ_0 is the minimum value of the vector θ and $\Psi = \Theta \otimes (\bar{A}P + P\bar{A}^\top + \beta P - c\bar{B}\bar{B}^\top) - \alpha\underline{\lambda}^\star_\Upsilon(I_N \otimes \bar{B}\bar{B}^\top) + I_{3N} + S + \sum_{k=1}^m (M_k + M_k^\top)$, being $\underline{\lambda}^\star_\Upsilon$ is the lower-bound of the minimum eigenvalue of the time-varying matrix $\Upsilon(t) = (\Theta\otimes 1)\bar{\Delta} b(t)\bar{b}^{-1}(\mathscr{H}\otimes 1)$. Then, choosing the PID-like control gains as

$$K^\top = \bar{B}^\top P^{-1}, \tag{27}$$

the robust exponential stability is achieved with a decay rate β and an attenuation index w.r.t. the parameter uncertainties equal to γ.

Proof: Consider the following Lyapunov function:

$$V(\xi(t)) = \xi^\top(t)(\Theta \otimes P^{-1})\xi(t). \tag{28}$$

Differentiating (28) along the trajectories of the closed-loop system in (24) and summing and subtracting the term $\beta V(\xi(t))$, with $\beta \in \mathbb{R}_+$, after several mathematical steps, we obtain:

$$\begin{aligned}\dot{V}(\xi(t)) + \beta V(\xi(t)) = {} & \xi^\top(t)\Big(\Theta \otimes (P^{-1}\bar{A} + \bar{A}^\top P^{-1})\Big)\xi(t) \\ & + \beta\xi^\top(t)\Big(\Theta \otimes P^{-1}\Big)\xi(t) \\ & + 2\xi^\top(t)\Big(\Theta \otimes P^{-1}\bar{B}\Big)\tilde{\Delta} f(t) \\ & - 2\alpha\xi^\top(t)\Big(\Theta\mathscr{H} \otimes P^{-1}\bar{B}\bar{B}^\top P^{-1}\Big)\xi(t) \\ & - 2\alpha\xi^\top(t)\Big(\Theta \otimes P^{-1}\bar{B}\Big)\bar{\Delta} b(t)\bar{b}^{-1} \\ & \times(\mathscr{H} \otimes \bar{B}^\top P^{-1})\xi(t).\end{aligned} \tag{29}$$

Introducing the state transformation $\tilde{\xi}(t) = (I_N \otimes P^{-1})\xi(t)$, the matrix $\Upsilon(t) = (\Theta \otimes 1)\bar{\Delta} b(t)\bar{b}^{-1}(\mathcal{H} \otimes 1)$ and applying Lemma 2, (29) can be recast as

$$\begin{aligned}\dot{V}(\xi(t)) + \beta V(\xi(t)) = {} & \tilde{\xi}^\top(t)\Big(\Theta \otimes \bar{A}P + P\bar{A}^\top + \beta P\Big)\tilde{\xi}(t) \\ & + 2\tilde{\xi}^\top(t)\Big(\Theta \otimes \bar{B}\Big)\tilde{\Delta} f(t) - \alpha\tilde{\xi}^\top(t) \\ & \times\Big(\Theta\mathscr{H} + \mathscr{H}^\top\Theta \otimes \bar{B}\bar{B}^\top\Big)\tilde{\xi}(t) \\ & - \alpha\tilde{\xi}^\top(t)\Big(I_N \otimes \bar{B}\Big)(\Upsilon(t) + \Upsilon^\top(t)) \\ & \times(I_N \otimes \bar{B}^\top)\tilde{\xi}(t).\end{aligned} \tag{30}$$

Applying Lemma 3 and the Rayleigh quotient on the symmetric term $\Upsilon(t) + \Upsilon^\top(t)$, we can recast (30) as

$$\begin{aligned}\dot{V}(\xi(t)) + \beta V(\xi(t)) < {} & \tilde{\xi}^\top(t)\Big(\Theta \otimes \bar{A}P + P\bar{A}^\top + \beta P\Big)\tilde{\xi}(t) \\ & + 2\tilde{\xi}^\top(t) \\ & \times\Big(\Theta \otimes \bar{B}\Big)\tilde{\Delta} f(t) - \alpha\underline{\lambda}_\Theta\theta_0\tilde{\xi}^\top(t) \\ & \times\Big(\Theta \otimes \bar{B}\bar{B}^\top\Big)\tilde{\xi}(t) \\ & - \alpha\underline{\lambda}_\Upsilon(t)\tilde{\xi}^\top(t)\Big(I_N \otimes \bar{B}\bar{B}^\top\Big)\tilde{\xi}(t),\end{aligned} \tag{31}$$

being $\underline{\lambda}_\Theta$ and $\underline{\lambda}_\Upsilon(t)$ the minimum eigenvalues of $(\Theta\mathscr{H} + \mathscr{H}^\top\Theta)$ and $(\Upsilon(t) + \Upsilon^\top(t))$, respectively. Finally, considering the lower-bound of $\underline{\lambda}_\Upsilon(t)$ as $\underline{\lambda}^\star_\Upsilon$ and introducing the positive parameter $c < \underline{\lambda}_\Theta\theta_0$, we obtain:

$$\begin{aligned}&\dot{V}(\xi(t)) + \beta V(\xi(t)) \\ &\quad < \tilde{\xi}^\top(t)\Big(\Theta \otimes \bar{A}P + P\bar{A}^\top + \beta P - c\bar{B}\bar{B}^\top \\ &\qquad - \alpha\underline{\lambda}^\star_\Upsilon(I_N \otimes \bar{B}\bar{B}^\top)\Big)\tilde{\xi}(t) + 2\tilde{\xi}^\top(t)\Big(\Theta \otimes \bar{B}\Big)\tilde{\Delta} f(t).\end{aligned} \tag{32}$$

In order to provide suitable robustness properties with respect to the vehicle parameter uncertainties, define the following performance index:

$$J = \int_0^{+\infty} (\tilde{\xi}^\top(s)\tilde{\xi}(s) - \gamma^2\tilde{\Delta}^\top f(s)\tilde{\Delta} f(s))ds, \tag{33}$$

being γ a positive attenuation index. Accordingly, following the approach in [43], the robust leader-tracking property is guaranteed if

$$\dot{V}(\xi(t)) + \tilde{\xi}^\top(t)\tilde{\xi}(t) - \gamma^2\tilde{\Delta}^\top f(t)\tilde{\Delta} f(t) < 0. \tag{34}$$

Therefore, given inequality (32), by adding and subtracting $\tilde{\xi}^\top(t)\tilde{\xi}(t) - \gamma^2\tilde{\Delta}^\top f(t)\tilde{\Delta} f(t)$ and introducing the vector $\zeta(t) = [\tilde{\xi}^\top(t), \tilde{\Delta}^\top f(t)]^\top \in \mathbb{R}^{4N\times 1}$, we obtain

$$\begin{aligned}&\dot{V}(\xi(t)) + \beta V(\xi(t)) + \tilde{\xi}^\top(t)\tilde{\xi}(t) \\ &\quad - \gamma^2\tilde{\Delta}^\top f(t)\tilde{\Delta} f(t) < \zeta^\top(t)\Pi\zeta(t).\end{aligned} \tag{35}$$

where the matrix $\Pi \in \mathbb{R}^{4N\times 4N}$ is defined as (36), shown at the bottom of the next page. Now, noting that, according to the Newton-Liebnitz formula [44], [45], $\forall\phi < 1$, it holds

$$\int_{\phi t}^{t} \dot{\tilde{\xi}}(s)ds = \tilde{\xi}(t) - \tilde{\xi}(\phi t), \tag{37}$$

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we leverage the Free Matrices method [46], i.e.

$$\Big[2\tilde{\xi}^\top(t)\sum_{k=1}^{m} M_k + 2\tilde{\xi}^\top(\phi t)T + 2\Big(\int_{\phi t}^{t}\dot{\tilde{\xi}}(s)ds\Big)^\top G\Big] \times\Big[\tilde{\xi}(t) - \tilde{\xi}(\phi t) - \int_{\phi t}^{t}\dot{\tilde{\xi}}(s)ds\Big] = 0. \tag{38}$$

being $M_k,\ k = 1, 2, \cdots, m$, with $m \in \mathbb{N}_{>0}$, T and G free matrices of suitable dimensions. Summing the null term (38) and adding/subtracting the term $\tilde{\xi}^\top(t)S\tilde{\xi}(t)$ to inequality (35), being S a positive definite matrix of suitable dimensions, we have:

$$\dot{V}(\xi(t)) + \beta V(\xi(t)) + \tilde{\xi}^\top(t)\tilde{\xi}(t) - \gamma^2\tilde{\Delta}f^\top(t)\tilde{\Delta}f(t) < \zeta^\top(t)\tilde{\Pi}\zeta(t) + \chi^\top(t)\Gamma\chi(t), \tag{39}$$

where $\chi(t) = \Big[\tilde{\xi}^\top(t),\ \tilde{\xi}^\top(\phi t),\ \Big(\int_{\phi t}^{t}\dot{\tilde{\xi}}(s)ds\Big)^\top\Big]^\top$ and

$$\Gamma = \begin{bmatrix} -S & -2\sum_{k=1}^{m} M_k + 2\ T^\top & -2\sum_{k=1}^{m} M_k + 2\ G^\top \\ 0 & -T - T^\top & -2T - 2G^\top \\ 0 & 0 & -G - G^\top \end{bmatrix},$$

$$\tilde{\Pi} = \begin{bmatrix} \Psi & \Theta\otimes\bar{B} \\ \star & -\gamma^2 I_N \end{bmatrix},$$

with $\Psi = \Theta\otimes(\bar{A}P + P\bar{A}^\top + \beta P - c\bar{B}\bar{B}^\top) - \alpha\underline{\lambda}^\star_\Upsilon(I_N\otimes\bar{B}\bar{B}^\top) + I_{3N} + S + \sum_{k=1}^{m}(M_k + M_k^\top)$. Choosing S, G and T as positive definite matrices we have that $\Gamma < 0$, due to its diagonal structure and, hence, inequality (39) can be recast as

$$\dot{V}(\xi(t)) + \beta V(\xi(t)) + \tilde{\xi}^\top(t)\tilde{\xi}(t) - \gamma^2\tilde{\Delta}f^\top(t)\tilde{\Delta}f(t) < \zeta^\top(t)\tilde{\Pi}\zeta(t) + \chi^\top(t)\Gamma\chi(t) \underset{\Gamma<0}{<} \zeta^\top(t)\tilde{\Pi}\zeta(t). \tag{40}$$

In so doing, if inequality (26) hols we obtain:

$$\dot{V}(\xi(t)) + \beta V(\xi(t)) + \tilde{\xi}^\top(t)\tilde{\xi}(t) - \gamma^2\tilde{\Delta}f^\top(t)\tilde{\Delta}f(t) < 0, \tag{41}$$

which leads to the following solution:

$$V(\xi(t)) < V(\xi(t_0))e^{-\beta(t-t_0)} - \int_{t_0}^{t} e^{-\beta(t-s)}\Big(\tilde{\xi}^\top(s)\tilde{\xi}(s) - \gamma^2\tilde{\Delta}f^\top(s)\tilde{\Delta}f(s)\Big)ds. \tag{42}$$

Applying the Rayleigh inequality [29] on the Lyapunov candidate function in (28), we can find the bound for the closed-loop state trajectories as

$$\|\xi\| < \Bigg(\frac{V(\xi(t_0))}{\underline{\lambda}_{(\Theta\otimes P^{-1})}}e^{-\beta(t-t_0)} - \frac{1}{\underline{\lambda}_{(\Theta\otimes P^{-1})}}\int_{t_0}^{t} e^{-\beta(t-s)} \times\Big(\tilde{\xi}^\top(s)\tilde{\xi}(s) - \gamma^2\tilde{\Delta}f^\top(s)\tilde{\Delta}f(s)\Big)ds\Bigg)^{\frac{1}{2}}, \tag{43}$$

being $\underline{\lambda}_{(\Theta\otimes P^{-1})}$ the minimum eigenvalue of the matrix $(\Theta\otimes P^{-1})$. Finally, evaluating (42) with zero initial conditions, i.e. $\xi(t_0) = 0$, from (43), it is simple to prove that

$$-\frac{1}{\underline{\lambda}_{(\Theta\otimes P^{-1})}}\int_{t_0}^{t} e^{-\beta(t-s)} \times\Big(\tilde{\xi}^\top(s)\tilde{\xi}(s) - \gamma^2\tilde{\Delta}f^\top(s)\tilde{\Delta}f(s)\Big)ds > 0, \tag{44}$$

which implies

$$\int_{t_0}^{t}\|\tilde{\xi}(s)\|^2 ds \le \gamma^2\int_{t_0}^{t}\|\tilde{\Delta}f(s)\|^2\ ds. \tag{45}$$

In so doing, the robust exponential stability with a decay rate β of the control protocol (20) is proven and this complete the proof. □

Remark 1: The LMI in (25)-(26) *is feasible and can be solved by using, for example, the interior-point method implemented in the Yalmip Toolbox with SeDuMi solver [47], [48].*

Remark 2: Theorem 1 provides an exponential stability condition expressed as the LMIs in (25) *and* (26). *Given a desired decay rate β for the closed-loop state trajectories, the solution of* (25) *and* (26) *allows the proper tuning of the control gains that ensures the desired transient behaviour and the convergence time for the longitudinal closed-loop trajectories.*

Remark 3: Note that, according to the theoretical derivation, our control design as in Theorem 1 consider desired convergence rates for the tracking of the optimal eco-driving profiles. This is a crucial aspect for our ecological application since it allows a precise leader-tracking with a desired transient behaviour, and hence, energy saving.

V. Performance Analysis

The effectiveness of both the robust exponential-stable PID-like and the Eco-Driving control architecture is verified in terms of both closed-loop and energy-consumption performance by carrying out extensive virtual simulations via the high-fidelity environment MiTRaS [26]. By way of example, a group of 5 CAEVs plus a leader, labeled with index 0, traveling as a convoy along a straight road is considered. The leading vehicle, driven by the NMPC control strategy (17), imposes the reference speed trajectory that the followers, equipped with the proposed distributed robust exponential-stable PID (20), have to track. As exemplar communication topology, we consider the L-P-F one. Nominal values of the heterogeneous longitudinal dynamics parameters, as well as the uncertainty range for each of them, are reported in Table I [49]. More specifically, to evaluate the performance of the platoon under the action of the proposed control architecture for all the different values of the unknown time-varying parameters uncertainties (modeled as random variables assumed to be uniformly distributed), as well as for all the possible combinations of them, we exploit Latin Hypercube Sampling (LHS) [50] approach. This latter allows generating random samples of the parameter with the aim of saving processing time when running Monte Carlo simulation. In so doing, a proper evaluation of the proposed

$$\Pi = \begin{bmatrix} \Theta\otimes\bar{A}P + P\bar{A}^\top + \beta P - c\bar{B}\bar{B}^\top - \alpha\underline{\lambda}^\star_\Upsilon(I_N\otimes\bar{B}\bar{B}^\top) + I_{3N} & \Theta\otimes\bar{B} \\ \star & -\gamma^2 I_N \end{bmatrix}. \tag{36}$$

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TABLE I
HETEROGENEOUS NONLINEAR VEHICLE PARAMETERS

Vehicle Parameters	$\bar{m}_j$ $[kg]$	$\bar{\eta}_j$ $[-]$	$\bar{R}_j$ $[m]$	$\bar{\mu}_j$ $[-]$	$\bar{C}_{D_j}$ $[-]$	$\bar{A}_{f_j}$ $[m^2]$
0	1545	0.89	0.3060	0.020	0.28	2.3315
1	1015	0.89	0.2830	0.022	0.30	2.1900
2	1375	0.89	0.2880	0.019	0.24	2.4000
3	1430	0.89	0.3284	0.021	0.28	2.4600
4	1067	0.89	0.2653	0.023	0.29	2.1400
5	1155	0.89	0.2880	0.024	0.33	2.0400
Uncertainty Range	$\pm 20\%$ $\bar{m}_j$	$\pm 5\%$ $\bar{\eta}_j$	- -	$\pm 10\%$ $\bar{\mu}_j$	$\pm 10\%$ $\bar{C}_{D_j}$	$\pm 10\%$ $\bar{A}_{f_j}$

eco-driving control architecture can be carried out. Regarding the battery pack parameters, instead, we consider, for each vehicle $j = 0, 1, \cdots, N$, the values reported in [38].

By selecting the decay rate $\beta = 0.5$, control gains for the distributed robust PID-like control strategy in (20) are computed according to Theorem 1 as: $K_P = 0.0347$, $K_I = 0.0347$, $K_D = 0.0868$, $\alpha = 21.8373$. Regarding the NMPC parameters, the weighing factors in (18) are chosen as: $\omega_1 = 10$; $\omega_2 = 10$; $\omega_3 = 20$; $\omega_4 = 20$; $\omega_5 = 0.1$; $\omega_6 = 10$. Since variables involved into the functional cost (19) have different magnitudes, proper scale factors (expressed in engineering units) are chosen according to the methodology proposed in [51] as: $s_1 = 2\ [s/m]$ for L_1; $s_2 = 172 \cdot 10^3\ [1/W]$ for L_2; $s_3 = 2\ [1/m]$ for L_3; $s_4 = 6\ [s^2/m]$ for L_4; $s_5 = 1\ [s/m]$ for L_5; $s_6 = 2\ [s^2/m]$ for L_6. Moreover, considering a sample time of 0.1 $[s]$ the NMPC prediction horizon is set to 0.5 s while the control horizon to 0.2 s.

A. Driving Scenarios and Key Performance Indexes

To disclose the effectiveness of the proposed Eco-Driving control architecture, we proceed in a twofold way. First, we prove its ability in guaranteeing the exponential convergence of the follower state trajectories to leader ones, as well as their boundedness property (according to inequality (43)) by considering the platoon formation manoeuvre. Then, the effectiveness of the control architecture in ensuring an ecological behaviour for the whole platoon is proven by considering the two following simulation scenarios: i) *trapezoidal speed profile*; ii) *realistic speed profile*. The *trapezoidal speed profile* mimics the effect of different traffic conditions, such as a sudden deceleration manoeuvre performed due to the presence of a forward obstacle (for instance, a vehicle not belonging to platoon or not connected) then followed by an acceleration to reach the target velocity as soon as it is possible [52]. Regarding the *realistic speed profile*, we consider the World harmonized Light-duty vehicles Test Procedure (WLTP) [53] as the reference speed profile to be followed. This driving cycle is divided into urban, suburban, rural and highway part, with an equal division between urban and non-urban paths (52% and 48% respectively) [54]. Each part is characterized by different average speeds, driving phases, acceleration and braking phases. Note that this profile is widely used for deepening the vehicles energy consumption performance. Moreover, as exemplar elevation profile, we consider the one of the A1 highway section connecting Naples to Capua.

TABLE II
PID CONTROL GAINS FOR DIFFERENT DECAY-RATE VALUES AND RELATIVE SETTLING TIMES

Decay rate	Control gains K_P	K_I	K_D	α	Settling time [s]
0.1	0.0203	0.0204	0.0417	14.9998	19.66
0.5	0.0347	0.0347	0.0868	21.8373	17.82
0.8	0.0347	0.0347	0.0868	21.8382	17.77
1	0.0345	0.0345	0.0860	21.8386	17.76

To carry out a quantitative assessment of the performance of the proposed Eco-Driving control architecture and to highlight the benefits it could bring to vehicles platoon, we consider the following Key Performance Indexes (KPIs): i) trend of the inter-vehicle profile $[m]$; ii) trend of the speed profile $[m/s]$; iii) Tracking Index (TI); iv) Energy Consumption (EC) $[kW/h\ km]$. The TI_i ($\forall i = 1, \cdots, N$), used to evaluate the leader tracking capability, is computed according to [32] as

$$TI_i = \frac{1}{T}\int_0^T \Big(|(\upsilon_i(t) - \upsilon_j(t)) \cdot SDE| + |(p_i(t) - p_j(t) - d_{ij}) \cdot SVE| \Big) dt \tag{46}$$

being T $[s]$ the simulation length while SVE and SDE are the sensitivity to the velocity error and distance error (see [55] for these values setting). Instead, the EC is evaluated for each vehicle j ($j = 0, 1, \cdots, N$), according to eq.(14).

VI. VIRTUAL SIMULATION RESULTS

A. Exponential Stability Validation: Platoon Formation Manoeuvre

In this section, we disclose the effectiveness, the robustness and the flexibility of the proposed PID control protocol (20) in ensuring, given a prescribed value for the decay rate β, the precise leader-tracking with a desired transient behaviour. To this end, we carry out different simulations involving the platoon formation maneuver for different values of the desired decay rate β and, for each of them, we compute the related control gains tuple by solving the LMIs as in (25)-(26) problems. In so doing, it is possible to confirm the feasibility of the LMI problem for different values of β, as well as to disclose the boundedness of the closed-loop state trajectories according to (43) for each appraised value of the decay rate.

For each simulation scenario we consider the same initial conditions for vehicles platoon positions and speed: [259.3, 239.1, 220.4, 200.7, 181.5] $[m]$ and [16, 14.5, 14, 15.5, 14] $[m/s]$ for followers; 280 $[m]$ and 15 $[m/s]$ for the leader. Simulations results are reported in Table II, where for each exemplar value of β is also associated the settling time the related computed control gains tuple can ensure. This relationship is also highlighted in Fig. 3, where it is shown how the settling time is greatly reduced when passing from $\beta = 0.1$ to $\beta = 0.5$, while its variation is exceedingly small for decay rate values $\beta \geq 0.8$. Finally, to further disclose the boundedness property of the closed-loop state trajectories according to inequality (43), we report, by the way of example, in Fig. 4 the evolution of the state trajectories norm over time for the case of $\beta = 0.5$.

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TABLE III

Trapezoidal Speed profile SCENARIO. LEADER TRACKING PERFORMANCES. COMPUTATION OF THE TI MEAN AND STANDARD DEVIATION

Control Approach	TI Statistics									
	1		2		3		4		5	
	Mean	Std	Mean	Std	Mean	Std	Mean	Std	Mean	Std
A) PID control proposed in [34]	0.356	0.043	0.563	0.064	0.729	0.078	0.602	0.046	0.588	0.045
B) Exponentially-Stable PID (20)	0.029	0.0018	0.040	0.0020	0.051	0.0023	0.049	0.018	0.051	0.0019
Reduction [%] (A-B)	91.94	91.87	92.89	93.79	93.08	94.16	91.88	92.09	91.32	91.33
C) NMPC+PID control proposed in [34]	0.23	0.028	0.35	0.041	0.45	0.049	0.37	0.028	0.36	0.027
D) Eco-Driving Control Architecture	0.015	0.0018	0.020	0.0020	0.025	0.0023	0.024	00.0018	0.025	0.0019
Reduction [%] (C-D)	93.62	93.50	94.35	95.04	94.47	95.34	93.50	93.69	92.99	92.92

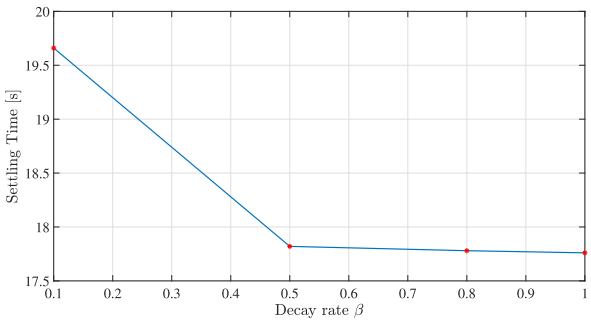

Fig. 3. Relation between the settling time $[s]$ and the decay rate β.

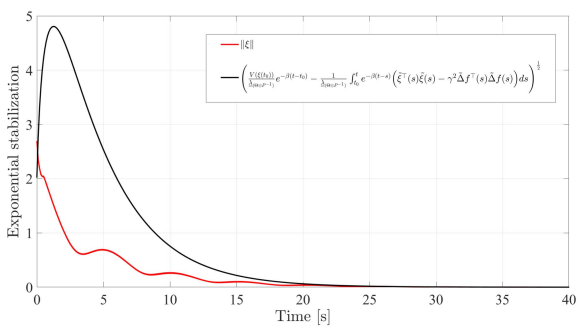

Fig. 4. Time history of the state trajectories norm vs time history of the state trajectories bound for the case $\beta = 0.5$.

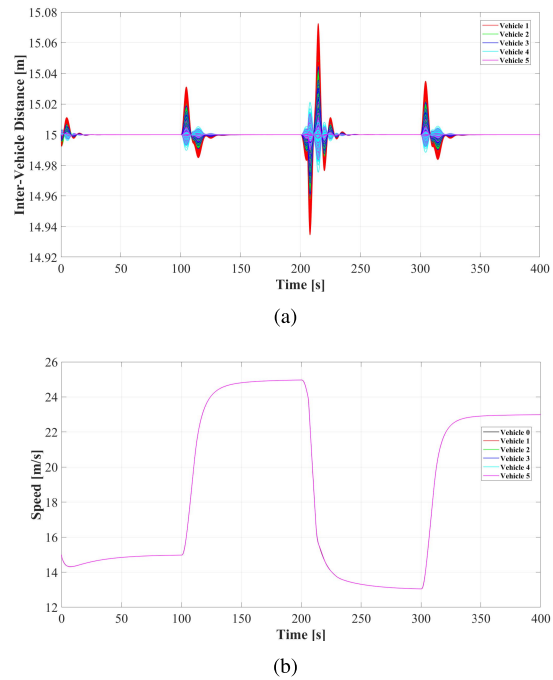

Fig. 5. *Trapezoidal Speed profile* scenario. Leader Tracking performance for the proposed Eco-driving Control Architecture. Robustness analysis via Latin Hypercube Sampling method. Time history of: a) vehicles inter-vehicle distance $[m]$; b) vehicles speed $[m/s]$.

B. Trapezoidal Speed Profile

Here, we evaluate the closed-loop performance of the proposed eco-driving control architecture for the *trapezoidal speed profile* scenario. The appraised profile is time-varying and composed of: 1) a speed constant phase of 15 $[m/s]$ for $t \in [0; 100[$; 2) an acceleration phase from 15 $[m/s]$ to 25 $[m/s]$ for $t \in [100; 110[$; 3) a speed constant phase of 25 $[m/s]$ for $t \in [110; 200[$; 4) a deceleration phase from 25 $[m/s]$ to 13 $[m/s]$ for $t \in [200; 210[$; 5) a speed constant phase of 13 $[m/s]$ for $t \in [210; 300[$; 6) an acceleration phase from 13 $[m/s]$ to 23 $[m/s]$ for $t \in [300; 310[$; 7) a speed constant phase of 23 $[m/s]$ for $t \in [310; 400]$.

1) Leader-Tracking Performance: Simulation results are reported in Fig. 5. As it is possible to observe herein, the Eco-Driving control architecture acts in a twofold way. First of all, the NMPC as in (17), embedded by the leader, allows the proper optimization of the trapezoidal speed profile (Fig. 5(b)). Then, the exponentially-stable distributed robust PID guarantees that each follower precisely tracks the leader motion with a desired transient behaviour and convergence time without exhibiting overshoot/undershoot (Fig. 5(b)) and with a small inter-vehicle distance error in the range $[-0.07; +0.07]$ $[m]$ (Fig. 5(a)), despite the presence of different uncertainties acting on vehicles dynamics.

To appreciate the solely advantages of the proposed PID controller in (20) in term of leader tracking performance we make a comparison between the following alternative Control Scheme (CS): i) CS_1, where both followers and the leader are equipped with the alternative distributed robust PID proposed in [34]; ii) CS_2, where both followers and the leader are equipped with (20). Simulation results, depicted in Fig. 6, prove how the proposed distributed robust exponential-stable PID has considerably improved leader-tracking performances w.r.t. the distributed PID in [34]. Indeed the controller proposed in [34] presents inter-vehicle distance errors into the range $[-1.15; 1.15]$ $[m]$ which are bigger than the one achievable via (20) (see Fig. 6(a) vs Fig. 6(c)) and exhibits overshoot/undershoot phenomena in the speed profile (see Fig. 6(b) vs Fig. 6(d)). In addiction, we also compute the TI for each vehicle i $(i = 1, \cdots, N)$ for both CS_1 and CS_2. Numerical results, reported in Table III, clearly prove how the proposed controller (20) allows enhancing the tracking performance of vehicles platoon with an average reduction for the whole fleet of about 92.22% for the mean value of the TI

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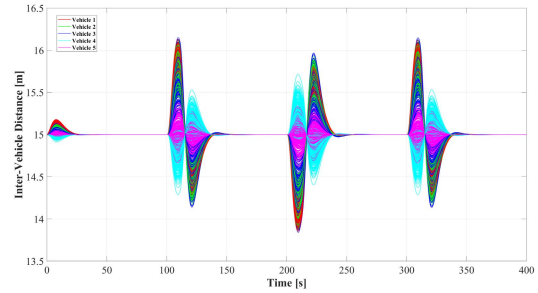
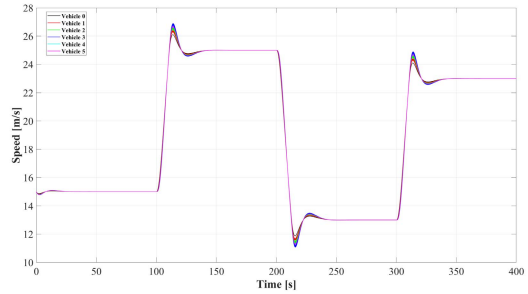


(a)



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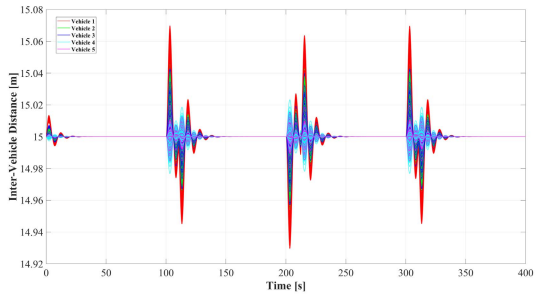

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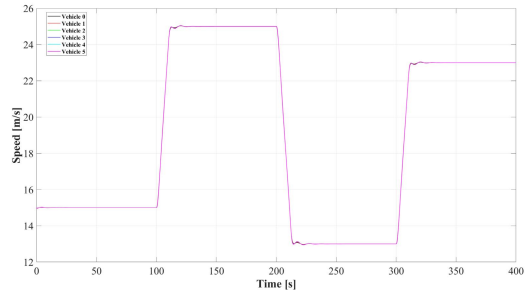
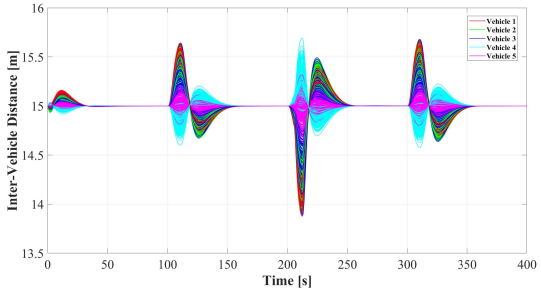


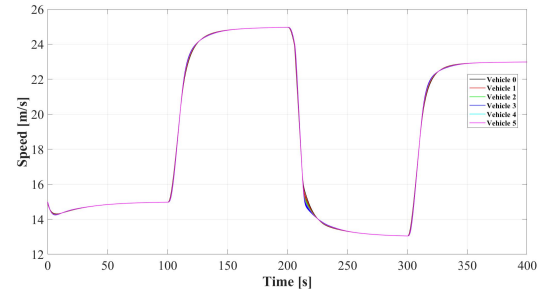
(d)

Fig. 6. *Trapezoidal Speed profile* scenario. Leader Tracking performance comparison between CS_1 and CS_2. Robustness analysis via Latin Hypercube Sampling method. Time history of: (a) inter-vehicle distance $[m]$ ensured by CS_1; (b) vehicles speed $[m/s]$ ensured by CS_1; (c) inter-vehicle distance $[m]$ ensured by CS_2; (d) vehicles speed $[m/s]$ ensured by CS_2.

and about 92.65% for its standard deviation. To further confirm the benefits of the proposed exponentially-stable controller in terms of leader-tracking, we also compare the Eco-Driving scheme with the alternative CS_3, where followers are equipped the controller in [34], while the leader with the NMPC as in (17). Leader tracking simulations for CS_3 are reported in Fig. 7 and, again, prove the superiority of the proposed exponentially-stable controller (20). Indeed, the inter-vehicle



(a)



(b)

Fig. 7. *Trapezoidal Speed profile* scenario. Leader Tracking performance for CS_3. Robustness analysis via Latin Hypercube Sampling method. Time history of: (a) inter-vehicle distance $[m]$; (b) vehicles speed $[m/s]$.

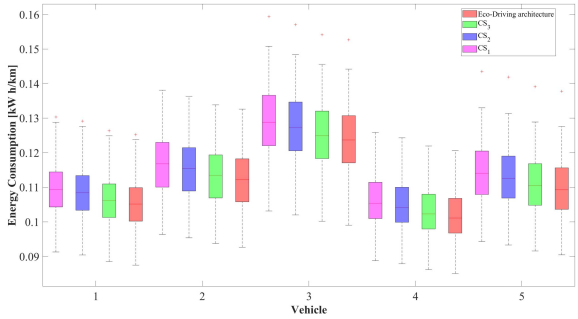


Fig. 8. *Trapezoidal Speed profile* scenario. Energy-saving performance. Boxplot Analysis. Comparison Analysis of the Eco-Driving control architecture w.r.t. control schemes CS_1, CS_2 and CS_3.

distance errors are in the interval $[-1.1; +0.6]\,[m]$ (see Fig. 7(a)) and the leader speed tracking in Fig. 6(b) is not so precise as the one ensured by the controller (20) (see Fig. 5(b)). This aspect is also highlighted by the computation of the TI index. Numerical results, reported in Table III, disclose how the eco-driving architecture allows obtaining an average reduction of about 93.8% for the mean value of TI and about 94.1% for its standard deviation w.r.t. CS_3.

2) Energy-Saving Performance: The main feature of the proposed eco-driving control architecture is the energy-saving consumption it can guarantee due to the combined effect of the optimization of the leader speed profile (guaranteed by the NMPC in (18)) and the precise leader tracking ensured by the proposed exponentially stable controller (20). To highlight this aspect, we compute the EC index for each vehicle i ($i = 1, \cdots, N$), hence obtaining the following average consumption: $[EC_1 = 0.1057,\ EC_2 = 0.1120,\ EC_3 = 0.1242,\ EC_4 = 0.1017,\ EC_5 = 0.1095]$. To better appreciate the energy-saving performance of the eco-driving architecture,

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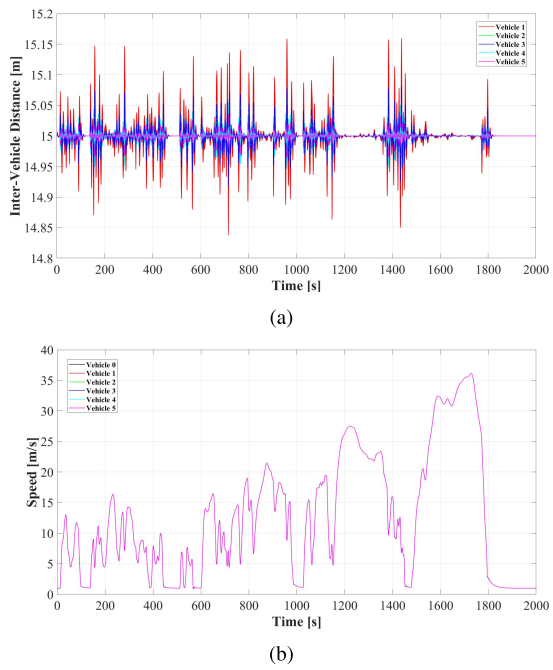

Fig. 9. *Realistic Speed profile* scenario. Leader Tracking performance for Eco-Driving Architecture. Robustness analysis via Latin Hypercube Sampling method. Time history of: (a) inter-vehicle distance $[m]$; (b) vehicles speed $[m/s]$.

we also compute the EC guaranteed by the different aforementioned Control Scheme, i.e. CS_1, CS_2 and CS_3, and compare them with the ones ensured by the eco-driving control scheme by leveraging the box-plot representation. Comparison results, depicted in Fig. 8 clearly highlight the benefit of the proposed architecture. More specifically, firstly, note how a first energy consumption reduction is guaranteed by leader optimization of the speed profile to be tracked. Indeed, the median value of control scheme CS_3 and eco-driving are lower than the ones of CS_1 and CA_2. Secondly, it is worth noting how the effect of the precise leader tracking ensured by (20) allows to further reduce the energy consumption of about [1.01%, 1.00%, 1.01%, 1.16%, 1.09%] w.r.t. CA_3.

C. Realistic Speed Profile

Here we evaluate the leader-tracking and energy-saving performance of the proposed eco-driving control approach for the *realistic speed profile* WLTP. To this aim, we also compare it with the one achievable via the alternative control scheme *baseline*, where the energy-saving purpose is not taken into account and leader/followers are driven by the solely control strategy (20). Note that, the comparison is carried out only w.r.t. *baseline* since our aim herein is to evaluate the advantages of the combined action of the NMPC (18) and the exponentially stable PID (20) in a realistic driving scenario. Finally, since our analysis is carried out for different values of the unknown time-varying parameters uncertainties via LHS method, statistical analysis is performed for the proper evaluation of the control approaches performance under the different appraised operating conditions [56].

1) Leader-Tracking Performance: Virtual simulation results, depicted in Fig. 9, confirm the ability of the proposed control approach in optimizing the speed profile to be tracked and ensuring that it is precisely tracked by the follower vehicles (Fig. 9(b)) with very bounded small inter-vehicle distance errors (Fig. 9(a)), despite the presence of parameters uncertainties. To evaluate the tracking capability of the proposed approach, we leverage the TI in (46) as KPI and we make a statistical comparison analysis with the baseline approach. To this aim, firstly, we exploit box-plots to display variations in samples of the statistical population of outputs (i.e. the variation of median, 25th and 75th percentiles) without making any assumptions of the underlying statistical distribution. Results in Fig. 10(a) highlight that the proposed eco-driving control approach improves the closed-loop performance of the whole platoon since both the median values (the central red mark) and the data dispersion are lower than the ones of the baseline approach. For the sake of completeness, we also report the mean values of TI and its percentage variation for each vehicle within the platoon in Table IV, hence observing a mean reduction of about 33.98% w.r.t. the baseline scenario. Then, histograms are exploited to represent the frequency distribution of TI of each vehicle for all the operative conditions. These statistical findings, comparing the shape of outputs distributions (Tracking Index (Fig. 10(b)-(f))), further confirm the benefits that each vehicle can achieve via the proposed control architecture w.r.t. the *baseline approach*: the reduction of the variance of the outputs distributions proves that the proposed approach is less sensitive to parameters uncertainty. Finally, TI distributions are compared each other to verify that obtained results are not due to the randomness of the experiment. Above all, the Lilliefors test [57] is performed to return a test decision for the null hypothesis that the data comes from a distribution in the normal family, versus the alternative that it does not come from such a distribution. We set the value of the significance level α (i.e., the probability of rejecting the null hypothesis when it is true) at 1%. The results of the test do not reject the null hypothesis except for the second vehicle. Once verified the normality of TI distribution, the two-sample t-test method [58] is used to test whether the population means of two groups are equal or not (for all vehicle but the second): the null hypothesis is that the data in the two samples come from independent normally distributed random samples with equal means and unknown unequal variances, while the alternative one is that the data come from populations with unequal means. For all vehicles cases, the means is verified to be significantly different at $\alpha = 1\%$ significance level; moreover, standard deviations are different with values of [0.0190, 0.0218, 0.0249, 0.0198, 0.0211] for baseline and [0.0127, 0.144, 0.0164, 0.0130, 0.0141] for eco-driving. Conversely, for vehicle 2, we perform the non-parametric

TABLE IV

Realistic Speed profile SCENARIO. LEADER TRACKING PERFORMANCES. COMPUTATION OF THE TI MEAN AND STANDARD DEVIATION

Control Approach	Mean TI				
	1	2	3	4	5
Baseline	0.15	0.22	0.27	0.26	0.28
Eco-Driving	0.10	0.14	0.18	0.17	0.18
TI Reduction [%]	33.61	33.88	34.41	34.27	33.72

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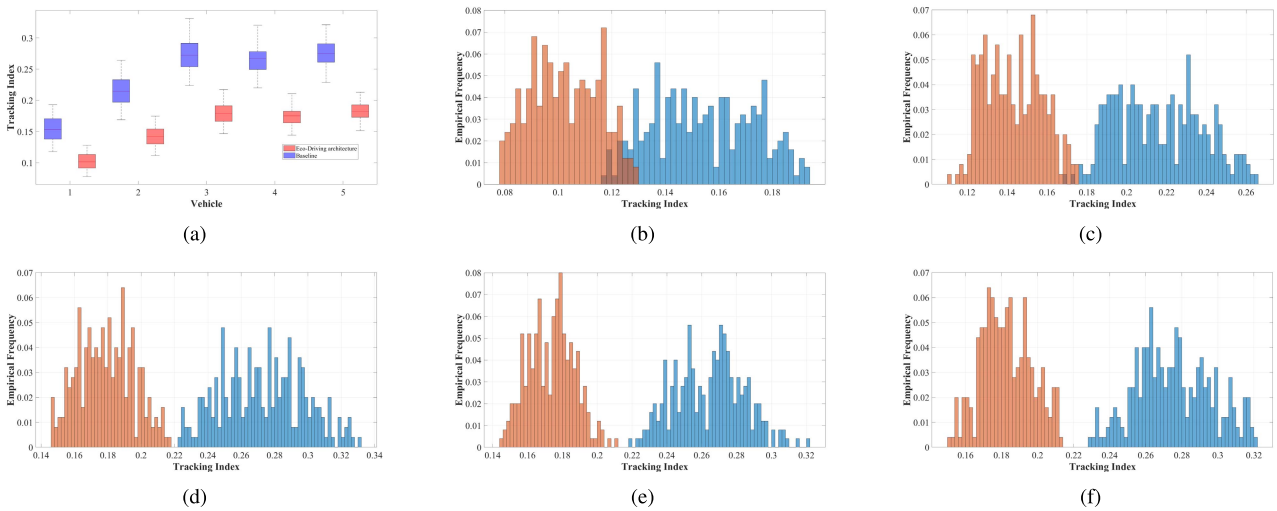

Fig. 10. *Realistic Speed profile* scenario. Leader Tracking performance. Robustness analysis via Latin hypercube sampling method. Tracking Index Analysis. Comparison between Eco-Driving approach and Baseline one: (a) box-plot of TI; (b) TI distribution for vehicle 1; (c) TI distribution for vehicle 2; (d) TI distribution for vehicle 3; (e) TI distribution for vehicle 4; (f) TI distribution for vehicle 5.

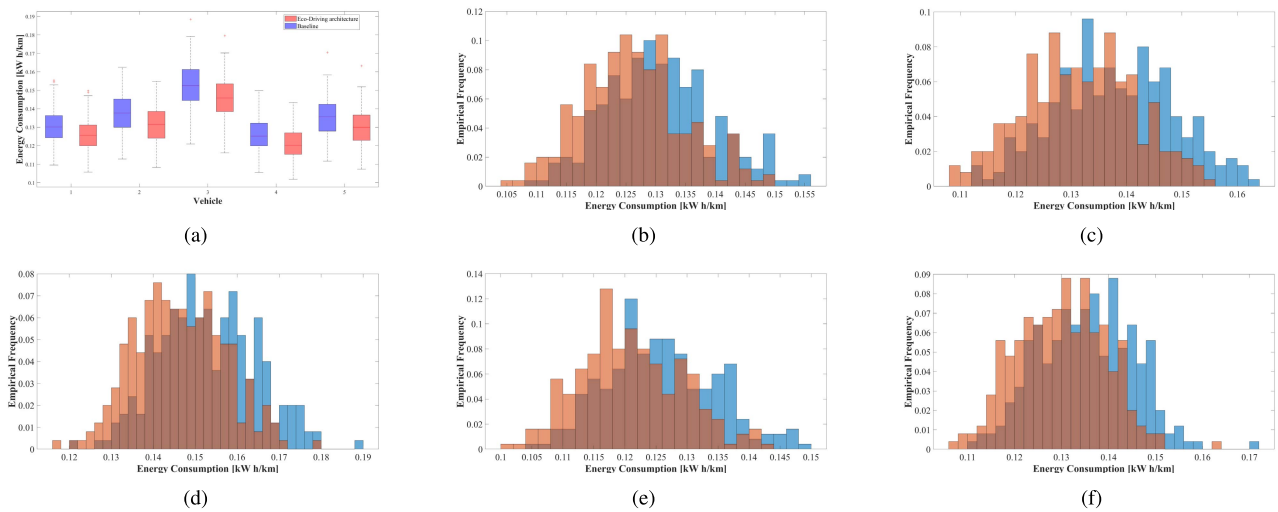

Fig. 11. *Realistic Speed profile* scenario. Energy-Consumption performance. Robustness analysis via Latin Hypercube Sampling method. Energy Consumption index. Comparison between Eco-Driving approach and Baseline one: (a) box-plot of EC; (b) EC distribution for vehicle 1; (c) EC distribution for vehicle 2; (d) EC distribution for vehicle 3; (e) EC distribution for vehicle 4; (f) EC distribution for vehicle 5.

Wilcoxon rank sum test [59]. The null hypothesis is that data are samples from continuous distributions with equal medians, against the alternative one that they are not. Testing outcome prove that the statistical distributions of the different samples are significantly different. Specifically, the null hypothesis (compared data comes from the same distribution) is rejected at $\alpha = 1\%$ significance level with $p - value = 2.2209e - 83$.

2) Energy-Related Results: To evaluate the energy-economy performance of the proposed eco-driving control architecture, we perform the same analysis as in VI-C-a. Box-plot results in Fig. 11(a) highlight that the proposed eco-driving control approach allows reducing the energy consumption for all the vehicles within the platoon. More specifically, both the median values and the data dispersion between the different parts of the box are lower than the ones of the baseline approach. Moreover, for the sake of completeness, in Table V we report the mean values of EC for each vehicle i with energy consumption reduction into the range [3.5; 4.6]% w.r.t. the baseline approach. These results,

TABLE V

Realistic Speed profile SCENARIO. ENERGY-CONSUMPTION PERFORMANCE. COMPUTATION OF THE EC MEAN VALUES AND ITS PERCENTAGE VARIATION

Control Approach	Mean Energy Consumption [kWh/km]				
	1	2	3	4	5
Baseline	0.131	0.138	0.153	0.126	0.135
Eco-Driving	0.126	0.132	0.146	0.121	0.130
EC Reduction [%]	3.573	4.537	4.404	3.906	3.975

combined with the reduction in standard deviation (with value [0.0090, 0.0106, 0.0114, 0.0088, 0.0100] for baseline approach and [0.0085, 0.0099, 0.0107, 0.0082, 0.0094] for eco-driving one), clearly indicates that vehicles adopt a smoother and present a reduced amount of dispersion. Despite consumption distributions having fairly similar shape, histograms in Fig.11(b)-(f) further confirm these results. Finally, EC distribution are compared each other to verify that obtained results are not due to the randomness of the experiment. Results of Lilliefors test do not reject the null

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hypothesis (data comes from a distribution in the normal family) at the $\alpha = 1\%$ significance level. Accordingly, we perform the two-sample t-test for each vehicle i within the platoon. For all the vehicles, the means have been verified to be significantly different at $\alpha = 1\%$ significance level. Again, also standard deviations are different with values of [0.0090, 0.0106, 0.0114, 0.0088, 0.0100] for baseline [0.0085, 0.0099, 0.0107, 0.0082, 0.0094] for eco-driving.

VII. Conclusion

This work proposes a novel distributed Eco-Driving control architecture for platoons of uncertain heterogeneous nonlinear CAEVs able to optimize the energy consumption performance while ensuring the fulfillment of the optimal leader trajectory tracking objective in a safe manner. The designed architecture consists of a NMPC strategy, driving the leader motion and optimizing in energy saving point of view the reference speed provided by the RSU, and a novel robust exponential-stable PID-like strategy, driving the follower motion so as to achieve the leader tracking performance with a desired prescribed decay rate. The exponential stability of PID-like controller has been proven via Lyapunov theory and the derived stability conditions have been expressed as a set of feasible LMIs, which allow the proper tuning of the control gains so to ensure the desired transient performances and the convergence time for the longitudinal closed-loop trajectories given the desired decay rate. Virtual simulations, carried in different operative scenarios and considering a wide range of values for the uncertain vehicles parameters via the virtual testing simulation platform MiTRaS, have confirmed the effectiveness of the proposed control Eco-Driving control architecture in achieving both energy-saving and leader-tracking control objectives.

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